WAIMLAp 2026

Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

Do Foundation Models Transfer Better Across Biomes? DINOv2/3 vs. U-Net for Forest Segmentation

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All 8 models hit the same wall: Amazon → Atlantic Forest ≈ 0.81 mIoU

DINOv2 and DINOv3 (web or satellite, up to 314 M parameters) gain at most ≈ 0.01 mIoU over an ImageNet U-Net; no gain is significant.

MOTIVATION & RESEARCH PROBLEM

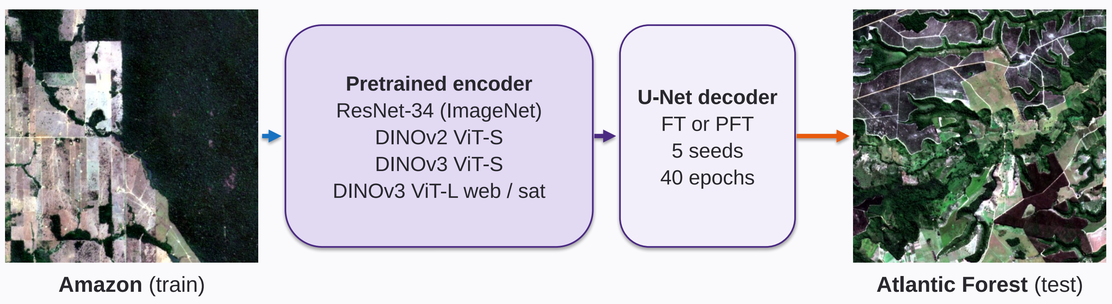
- Forest maps from Sentinel-2 must also work in **regions the model has never seen**.
- Inside one biome the task is **saturated**: U-Nets reach ≈ 0.95 IoU on the Amazon [2], [3], so in-biome benchmarks cannot rank encoders.
- Across biomes it is not: trained on the Amazon, a U-Net reaches only ≈ 0.81 on the Atlantic Forest [3].
- Self-supervised foundation models (DINOv2 [4], DINOv3 [5]) promise general features; DINOv3 even has a satellite-pretrained version.

Question: do DINOv2 / DINOv3 encoders transfer better from the Amazon to the Atlantic Forest than an ImageNet U-Net, and does satellite pretraining help?

OBJECTIVES / CONTRIBUTION

- **Controlled backbone swap:** 8 encoder / fine-tuning variants with the same U-Net decoder, data, schedule and seeds.
- **Cross-biome stress test** in both directions, including the whole unseen biome (605 / 619 images, 30× the official test set).
- **Paired statistics** over 5 seeds (80 runs); open code, result files and checkpoints.
- **Honest negative result:** what did not work.

METHODOLOGY



Tested on the official test split (20 images) and on the whole unseen biome (605 images). The reverse direction (Atlantic Forest → Amazon) is run as well.

- **Data:** Bragagnolo et al. [1], Sentinel-2 L2A, 10 m, bands R, G, B, NIR, 512 × 512 px; Amazon 499 / 100 / 20 and Atlantic Forest 485 / 100 / 20 images (train / val / test); the biomes share no tile.
- **Encoders:** ResNet-34 (ImageNet), DINOv2 ViT-S/14, DINOv3 ViT-S/16, DINOv3 ViT-L/16 web (LVD-1689M) vs. satellite (SAT-493M, 0.6 m RGB); one U-Net decoder (UNETR pyramid for ViTs).
- **Fine-tuning:** full (FT) or partial (PFT: first layer, last ViT blocks or ResNet stage 4, decoder); ViT-L PFT only.
- **Training:** 40 epochs of random 256 px crops with flips and rotations, AdamW + cosine, BCE + Dice; checkpoint chosen on the training biome's validation split only.
- **Statistics:** 5 seeds paired by seed, paired t-tests with Holm correction; main metric mIoU.

CONCLUSIONS

- Foundation-model encoders (DINOv2, DINOv3, web or satellite, up to 314 M parameters) **gain at most ≈ 0.01 mIoU** over an ImageNet U-Net from the Amazon to the Atlantic Forest; no gain is significant.
- In-biome tests are saturated and partly leaky (5 of 20 Atlantic test images repeat training ground): **cross-biome tests** are needed to compare encoders.
- The shared errors suggest that the limit is the **biome shift** (look-alike land cover, label definitions), not the features.
- **Limitations:** one dataset and one pair of biomes, 20 official test images per biome, 5 seeds, one training schedule, 4 bands; SAT-493M is 0.6 m RGB, our input 10 m with NIR.
- **Next:** domain adaptation, multi-biome training, a label audit, Sentinel-1.

Take-home: bigger or satellite-pretrained encoders did not fix biome shift.

RESULTS & DISCUSSION

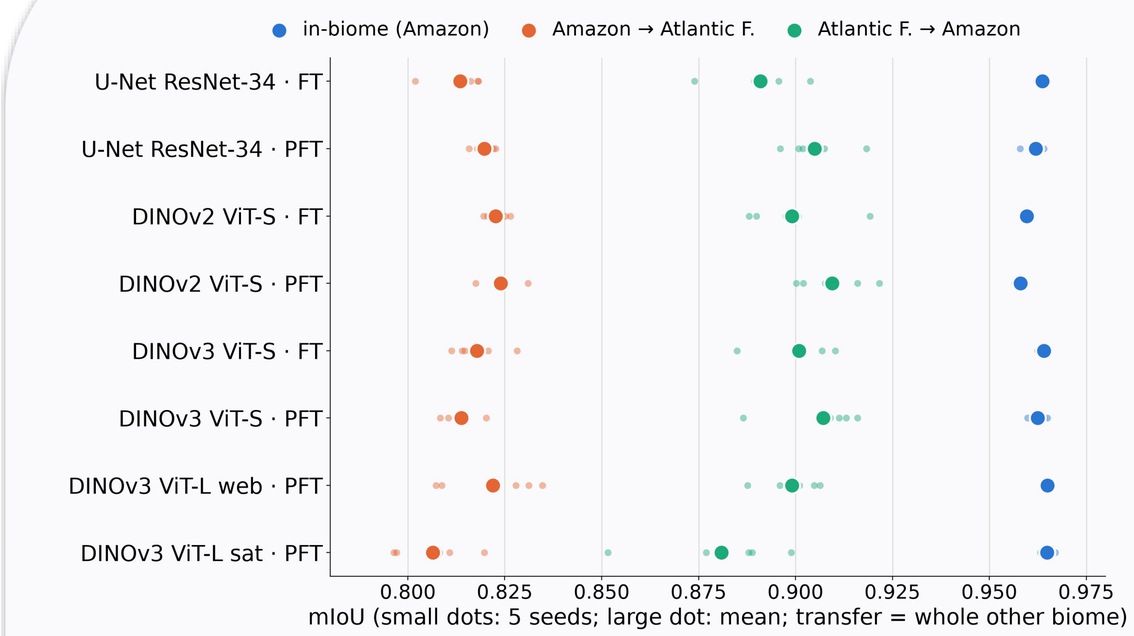


Fig. 1. mIoU of all 8 models in three settings (5 seeds each). Every model drops to ≈ 0.81 on the Atlantic Forest; the encoder barely moves it.

- **In-biome is saturated:** 0.958–0.965 mIoU on the Amazon, 0.923–0.933 on the Atlantic Forest, for all models.
- **Amazon → Atlantic Forest: 0.806–0.824 for every model** (whole biome; weighted IoU 0.82–0.84 vs. ≈ 0.81 in [3]). The reverse direction transfers much better (0.88–0.91).

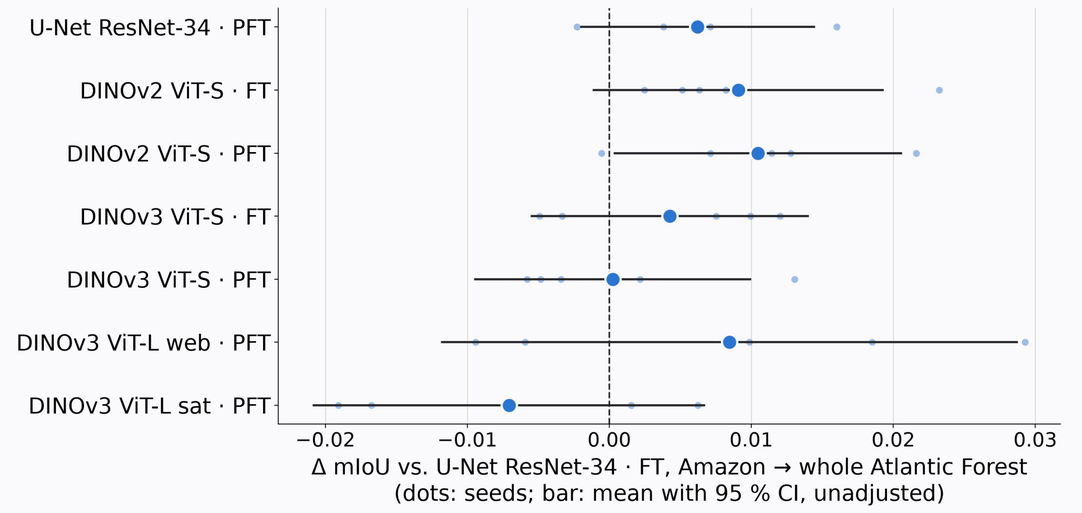


Fig. 2. Paired gain over the U-Net (FT) on the whole Atlantic Forest. The largest is +0.010 mIoU (DINOv2 ViT-S PFT); after Holm correction over the 7 comparisons none is significant (all adjusted p ≥ 0.31).

- **Satellite pretraining does not help:** ViT-L sat equals ViT-L web in-biome (0.965) but is 0.016 lower on transfer (better in 1 of 5 seeds, not significant), at 4.6× the trained parameters and 6× the training time of the U-Net.

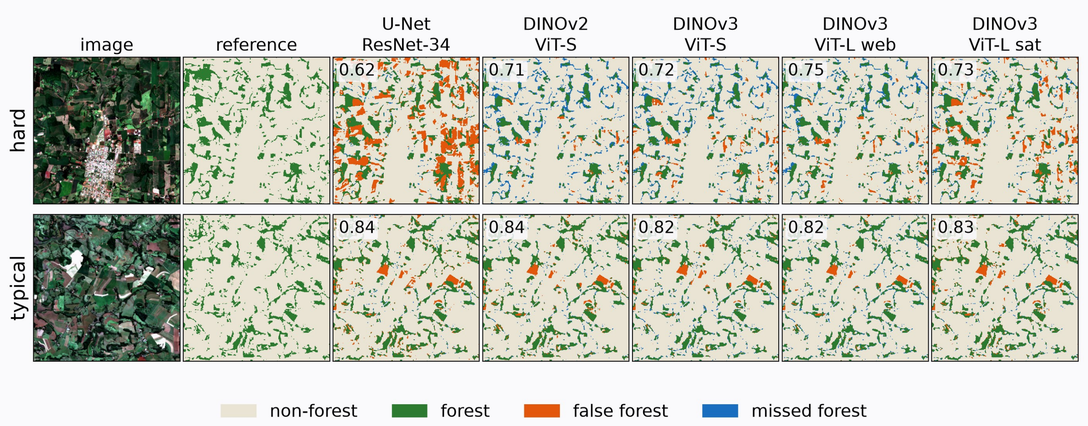


Fig. 3. Amazon-trained models on two Atlantic Forest test images (seed 0; U-Net and ViT-S: FT; hard / typical = 10th / 50th percentile of the image mIoU; number = image mIoU). All models also mark the same crop fields as forest (orange).

- **Errors are mostly false forest:** precision 0.81–0.85 vs. recall 0.90–0.91 (whole biome); the forest area is over-estimated by 2–4 points.

KEY REFERENCES

[1] L. Bragagnolo, R. V. da Silva, and J. M. V. Grzybowski, “Amazon and Atlantic Forest image datasets for semantic segmentation,” Zenodo, 2021, doi:10.5281/zenodo.4498086.

[2] L. Bragagnolo, R. V. da Silva, and J. M. V. Grzybowski, “Amazon forest cover change mapping based on semantic segmentation by U-Nets,” Ecol. Inform., vol. 62, p. 101279, 2021.

[3] D. John and C. Zhang, “An attention-based U-Net for detecting deforestation within satellite sensor imagery,” Int. J. Appl. Earth Obs. Geoinf., vol. 107, p. 102685, 2022.

[4] M. Oquab et al., “DINOv2: Learning robust visual features without supervision,” Trans. Mach. Learn. Res., 2024.

[5] O. Siméoni et al., “DINOv3,” arXiv:2508.10104, 2025.

ACKNOWLEDGMENTS

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CODE & DATA

Code, results (CSV) and checkpoints:
github.com/dsbferris/WAIMLAp-2026
Contact: c.calderonb@pucp.edu.pe

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